Zach Feldman

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EDUCATION

Cornell University College of Engineering, Ithaca, NY **August 2023 - May 2027 (Expected)**

- Bachelor of Science in Mechanical Engineering, Minor in Education
- **Relevant Coursework (*currently enrolled):** Dynamics, Mechanical Design, Nanoscience & Nanoengineering, Thermodynamics, Statics, System Dynamics, Mechanics of Materials, Fluid Dynamics, Heat Transfer*, Mechatronics*

EXPERIENCE

Cornell Electric Vehicles **October 2023 - Present**
Chassis Team Lead

- Collaborating with 85+ team members of varying disciplines to create hyper-efficient, autonomous electric cars, coming in 5th on the track and winning two Off-track awards ($4.5K) from Shell Eco-marathon Americas competition.
- Lead a team of 8, delegating tasks and overseeing workflow management to ensure all chassis projects are completed on schedule and seamlessly integrated with the broader team's deliverables.
- Optimize mold preparation and carbon fiber manufacturing by refining surface prep, polishing, release methods, layup procedures, and infusion parameters, reducing manufacturing time by ~50% while improving part strength and repeatability.
- Oversee vacuum-infusion and wet-layup composite manufacturing operations, manufacturing 8 high-quality carbon-fiber components across structural and aerodynamic vehicle body surfaces.
- Developed and manufactured 3D-printed headlights and taillights for a race car; completed 17 design iterations, converging on a final design while ensuring electronic compatibility and structural stability.
- Oversee procurement of build materials and assembly equipment within a $60k budget, optimizing inventory processes and coordinating with suppliers to streamline production workflows.

Cornell Extreme Mechanics, Materials, and Manufacturing (EM3) Lab **January 2026 - Present**
Undergraduate Researcher

- Investigating laser powder bed fusion (LPBF) process parameters and manufacturing strategies through experimental builds and post-process evaluation to assess their impact on material strength and overall part performance.

Revolutionary Cooling Systems **May 2025 - August 2025**
Mechanical Engineering Intern

- Designed the primary housing and integrated fluid-handling components for a new commercial beverage chilling product, encompassing enclosures, piping, spray nozzles, reservoirs, and mounts.
- Developed and manufactured ~50 new parts to support ongoing R&D projects using SolidWorks and various 3D printers.
- Spearheaded the company's 3D printing operations, expanding from 1 outdated printer to a 4-printer, full-capacity additive manufacturing lab; owned machine selection, scheduling, maintenance, and workflow optimization while advising the CEO on advanced printer procurement.
- Kept regular documentation, delivered several presentations, created multiple standard operating procedure documents, and authored a 50-page final report.

Cornell Athletics **September 2023 - Present**
Course Instructor & Lifeguard

- Teach beginning swim since September 2025, helping students build skills to pass Cornell's required swim test.
- Lifeguard for Cornell PE classes, the swim team, and recreational swimmers, providing instruction and advice as needed.

Mohawk Day Camp **June 2024 - August 2024**
General Counselor

- Supervised and mentored a group of 15 kindergarten-aged campers, maintaining a safe and supportive environment while collaborating with counselors and directors to coordinate daily programming and activities.

PROJECTS

Automatic Basketball Return System

- Engineered a modified basketball mini-hoop system that automatically returned balls to the user by integrating motors, 3D-printed flywheels, a laser-cut acrylic casing, and Arduino-controlled electronics.
- Collaborated with a 5-member engineering team to design, prototype, and manufacture the system.

3D Modeling/Printing Personal Project

- Modeled and 3D-printed a 2+ ft 1/128-scale Falcon 9 rocket from scratch using Fusion 360 and an IDEX 3D printer.
- Iterated on the design by running dozens of 3D printing tolerance and dual-extrusion tests to ensure proper component fitment.

Electronic Medical Device Personal Project

- Developed a prototype to motorize an endotracheal tube to be used at hospitals using an Arduino, servos, and coils so that a doctor could adjust the position of the tube easily while in someone's throat.
- Received direct feedback from EMTs and doctors to iterate and refine the design.

SKILLS

Technical: Autodesk Fusion 360, Autodesk Inventor, SolidWorks, ANSYS Mechanical, 3D Printing Slicers, DFM/DFA, Engineering Drawings, Arduino, LaTeX, Rapid Prototyping, Testing & Validation

Manufacturing: Composites (Carbon Fiber Wet Layup/Vacuum Infusion), 3D Printing, Laser Cutting, Mill, Lathe, Hand Tools